Week 4

Course Level Objectives:

- Recognize, examine, and interpret the basic principles of describing and presenting data. (CLO 2)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Describe the shape of a distribution from a dotplot or histogram. (CLO 2) **2.** Compute the mean and median, and explain each one's resistance to skewness and outliers. (CLO 2) **3.** Interpret the standard deviation as the typical distance of data values from the mean. (CLO 2) **4.** Use the 95% rule to estimate the standard deviation from a bell-shaped dotplot. (CLO 2)	e-Text: 2.2 One Quantitative Variable Interactive Lecture: 2.2 One Quantitative Variable Slides: 2.2 One Quantitative Variable	Participation: Quantitative Variables – shape, mean vs. median Your Turn: Mean vs. Median with an Outlier (commute times) Your Turn: Estimation Activity – estimate the mean and SD of a dotplot without calculating (95% rule) Participation: One Quantitative Variable (Kahoot!) – Wed: review of 2.2	Online Homework: 2.2 One Quantitative Variable (due Sun Sep 20, 11:59 PM) Take-Home Quiz 2 (due Mon Sep 21, 11:59 PM) – covers 2.1–2.3 (all objectives) Test 1 (Fri Oct 2 – Sun Oct 4, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Oct 1) (all objectives)	WileyPLUS StatKey Blackboard Kahoot! Respondus
5. Compute the sample standard deviation of a small dataset by hand. (CLO 2) **6.** Compute and interpret z-scores to compare values on different scales or in different groups. (CLO 2) **7.** Find the five-number summary, range, and IQR of a dataset. (CLO 2) **8.** Apply the 1.5 × IQR rule to identify outliers. (CLO 2) **9.** Interpret a boxplot to describe a distribution's center, spread, skewness, and outliers. (CLO 2)	e-Text: 2.3 One Quantitative Variable: Using Percentiles Interactive Lecture: 2.3 One Quantitative Variable: Using Percentiles Slides: 2.3 One Quantitative Variable: Using Percentiles	Participation: Using Percentiles – five-number summary Your Turn: Comparing Across Groups (heights of men and women with z-scores)	Online Homework: 2.3 One Quantitative Variable: Using Percentiles (due Sun Sep 20, 11:59 PM) Take-Home Quiz 2 (due Mon Sep 21, 11:59 PM) – covers 2.1–2.3 (all objectives) Test 1 (Fri Oct 2 – Sun Oct 4, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Oct 1) (all objectives)	WileyPLUS StatKey Blackboard Respondus